Highly Improved Staggered Quarks

D. A. Clarke (HotQCD, Fermilab-MILC)

A HISQ-smeared link U^{HISQ} is computed from a thin link U by [1]

$$U^{\text{HISQ}} = \mathcal{F}^{\text{corr}}\mathcal{U}\mathcal{F}^{\text{Fat7}}U, \tag{1}$$

i.e. we Fat7-smear [2] the thin link, followed by a reunitarization, and then a "corrected" AsqTad [2] smear, which retunes the one-link and Naik coefficients

$$c_1 = 1 + \frac{\epsilon}{8}, \qquad c_{\text{Naik}} = -\frac{1}{24}(1+\epsilon). \tag{2}$$

How to choose ϵ depends on the quark mass; for more details see eq. (24) of Ref. [1].

References

[1] E. Follana et al. Highly improved staggered quarks on the lattice with applications to charm physics. *Phys. Rev. D*, 75(5):054502, 2007.

[2] K. Orginos, D. Toussaint, and R. L. Sugar. Variants of fattening and flavor symmetry restoration. *Phys. Rev. D*, 60(5):054503, 1999.